

GitHub Implementations — Expanded Curation

Five research-connected implementations/benchmarks for tool selection and evaluation

1. How projects were selected

The assignment recommends examining documentation, source availability, activity, examples, reproducibility, license and research connection rather than choosing repositories only by star count. The five projects below are selected for direct relevance to tool-use evaluation.

1. ToolBench / ToolLLM

GitHub: <https://github.com/OpenBMB/ToolBench>

What it implements: Large-scale tool-learning ecosystem with APIs, data and evaluation-related components.

Why relevant: Strong match for large candidate-tool pools and tool-use research.

2. AnyTool

GitHub: <https://github.com/dyabel/AnyTool>

What it implements: Hierarchical API retrieval, solving and self-reflection for large-scale API calls.

Why relevant: Directly relevant to retrieval and tool selection at scale.

3. RoTBench

GitHub: <https://github.com/Junjie-Ye/RoTBench>

What it implements: Benchmark/code/data for robustness in tool learning.

Why relevant: Directly evaluates tool selection under noise.

4. ToolTalk

GitHub: <https://github.com/microsoft/tooltalk>

What it implements: Conversational tool-use benchmark with reference tool sequences.

Why relevant: Useful for sequence-level selection evaluation.

5. UltraTool

GitHub: <https://github.com/JoeYing1019/UltraTool>

What it implements: Research implementation associated with comprehensive tool-utilization benchmarking.

Why relevant: Useful for planning, tool creation and usage in complex scenarios.

2. Reproducibility checklist

- Repository can be accessed publicly
- README explains setup
- Code or data is present
- License is visible
- Examples or benchmark instructions exist
- Research paper/project connection is identifiable

■ The implementation can be mapped to a repository research category

3. Suggested comparison table

Project	Selection focus	Evaluation focus	Best use
ToolBench	Large API/tool pool	Tool-use tasks	Large-scale experiments
AnyTool	Hierarchical retrieval	API selection/calling	Large tool libraries
RoTBench	Robust selection	Noise robustness	Robustness experiments
ToolTalk	Conversation sequences	Tool sequence correctness	Multi-turn tasks
UltraTool	Planning/tool usage	Comprehensive capability	Complex scenarios